

CE 524 : Applied Hydroclimatology

Course Code : CE 524

Course Name : Applied Hydroclimatology

L-T-P-C : 3-0-0-3

Intended for : PG and Ph.D. students

Prerequisite : None; however, familiarity with any computer programming language (e.g., Python, R, MATLAB, etc.) will be helpful during the course.

Mutual Exclusion: None

Approval: 56th BoA

Course Contents

- **Introduction to hydroclimatology Climate change:** causes and impacts, Impact on global hydrologic cycle; Climate feedback, Tipping points, Water management in changing climate; Concept and scope of hydroclimatology. (4 Lectures)
- **Module II:** Observation Types of research data; Qualitative and quantitative data; Measurement of precipitation, temperature, snow, streamflow, snow water equivalent; Missing data handling techniques; Hydroclimatic measurement network in the Himalayan regions. (4 Lectures)
- **Module III:** Modeling Fundamental principles in hydroclimatic modeling; Types of climate models; Simple climate models, Simple energy balance climate models (e.g., zero- dimensional, single-layer, and one-dimensional); General circulation model (GCM); Hydrologic modeling; types of hydrological models; Components of hydrologic models; SWAT model; VIC model; HEC-HMS.
- **Module IV:** Hydroclimatic data processing and analysis Gridding observed data; Reanalysis data: Satellite-derived data; Radar data; Climate data; Paleoclimatology data; Climate indices, Overview of CMIP6; Downscaling and bias correction of climate data; Impact assessment studies. (8 Lectures)
- **Module V:** Impact on hydrology, agriculture, and ecosystem Climate change impact on Dams; Weather and climate metrics for agriculture, Heat stress indices; Impact of climate change on food security in India; Energy limitation versus moisture limitation; Ecosystem services; Himalayan ecosystem and impact of climate change. (6 Lectures)
- **Module VI:** Hydroclimatic extremes: Droughts and Floods Drought definition: meteorological droughts, hydrological droughts, agricultural drought, socioeconomic drought; Drought indicators: Theory of run; Severity; Duration; Intensity; Frequency analysis; Impact of climate change on droughts; (6 Lectures)
- **Module VII:** Flood types; Extreme precipitation indices; Flood analysis; Flood Frequency analysis; Design Flood; Flood hazard and damage; Glacial lake outburst flood (GLOF); Impact of climate change on floods; Flood risk management and floodplain management. (6 Lectures)

Text Books:

1. Shelton, M. L. , Hydroclimatology: perspectives and applications. Cambridge University Press, 2009.
2. VenTe, C., Maidment, D. R., & Mays, L. W., Applied hydrology, 1988.



Reference Books:

1. Beven, K. J., **Rainfall-runoff modelling: the primer**, John Wiley & Sons, 2011.
2. McGuffie, K., & Henderson-Sellers, A., **The climate modelling primer**, John Wiley & Sons, 2014.